Assignment — Land-cover patterns and possible futures in your commune

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Introduction

In the tutorial, you explored three ways of understanding a landscape. You interpreted satellite images by eye, trained a Random Forest to classify land cover, and used SSP-CH scenarios to explore possible future land-use patterns. Now apply these approaches to your assigned commune.

Your task is to explain what these methods reveal about your commune and how confidently you can interpret the results. Use the figures as evidence and connect your findings to the commune's landscape, settlement pattern and local conditions.

Scope of the report

Submit a report of **no more than two pages, including figures and captions**. Present the four figures below, and discuss the findings. Integrate your interpretation with the figures to use the available space effectively.

Use **2025 as the present-day reference and a year of your liking as the future time horizon**, following the tutorial. Apply the same commune boundary throughout. Give every figure a clear caption identifying the data, year and scenario where relevant.

Required figures

1. Interpret the landscape by eye

Create **one figure with two panels** showing natural-colour satellite images of your commune in **2018 and 2025**. Use the same map extent and display settings for both years.

Describe the main landscape features and identify an area that appears stable and an area where change is possible. Explain the evidence for your interpretation and where you are uncertain. Consider that 2018 was a drought.

2. Classify land cover automatically

Create **one figure showing the 2025 Random Forest land-cover prediction** for your commune. Follow the tutorial by fitting the model using the supplied reference data, predicting over the full raster extent, and then cropping the prediction to your commune.

Report the **overall accuracy and balanced accuracy** from the reserved test observations. Explain what these statistics tell you, and inspect the map alongside the satellite image. Discuss one convincing result and one questionable classification, using what you learned about reference labels, mixed pixels and confusion between classes.

3. Explore two possible futures

Choose **two SSP-CH scenarios** and create **one figure with four panels** showing land cover in **2025 and the future for each scenario**. Arrange one scenario per row and one year per column. Use consistent class colours and legends.

Explain why you selected these scenarios and which of their assumptions matter for your commune. Describe where their projected land-use patterns differ and relate those differences to the local landscape. Consider, for example, the distribution of settlements, agricultural land and forest.

4. Quantify change within each scenario

Create **one figure with two panels**, with one panel for each selected scenario. Show the change in area of each land-use class between 2025 and 2080.

For each class, calculate

$$\Delta A = A_{future} - A_{2025}.$$

Positive values indicate an area gain and negative values indicate an area loss. State the units, use the same axis scale in both panels, and use the same class colours as in the maps.

This figure should show change through time within each scenario. Compare the panels to explain how the two scenarios differ in the direction and magnitude of those changes.

Discussion and local context

Connect the results to the questions raised in the tutorial.

- **Manual interpretation.** Which features could you recognise confidently? What made identifying change difficult, and what additional evidence would help?
- **Automatic classification.** How did the Random Forest help you describe the landscape? Which errors or limitations matter most for interpreting the results in your commune?
- **Future scenarios.** Which assumptions produce the differences you observe? How do the maps and the area-change plots complement one another?
- **Your commune.** How do its physical setting, current land uses and settlement pattern help explain the results? Where do the projections appear plausible, and where would local knowledge be needed to assess them?

Support your statements with specific examples from your figures, e.g. by referencing them: (Fig. 1). Keep the uncertainty of your image interpretation and classification in view when discussing the present landscape based on the RandomForest. Explain future outcomes in relation to the assumptions of the selected scenarios.

End with one main finding about your commune and one important uncertainty or question that remains.